

CSE370 : Database Systems

Assignment 01/02/03 | Spring 2026

ID : <8 Digit Student ID> | Name : Full Official Name

No 1 Query (as Plain Text)	<pre>CREATE TABLE employees (-> e_id CHAR(4), -> first_name VARCHAR(20), -> last_name VARCHAR(20), -> hire_date DATE, -> job_id CHAR(10), -> salary INT, -> commission_pct DECIMAL(5,3), -> manager_id CHAR(4), -> dept_id CHAR(3) ->);</pre>
No 1 SS (of Query & Output in Shell)	<pre>MariaDB [Company_23201376]> CREATE TABLE employees (-> e_id CHAR(4), -> first_name VARCHAR(20), -> last_name VARCHAR(20), -> hire_date DATE, -> job_id CHAR(10), -> salary INT, -> commission_pct DECIMAL(5,3), -> manager_id CHAR(4), -> dept_id CHAR(3) ->); Query OK, 0 rows affected (0.120 sec)</pre>
No 2 Query (as Plain Text)	<pre>INSERT INTO employees (e_id, first_name, last_name, hire_date, job_id, salary, commission_pct, manager_id, dept_id) VALUES</pre>

	('E001', 'John', 'Doe', '2020-01-15', 'IT_PROG', 65000, NULL, 'E010', 'D01'), ('E002', 'Jane', 'Smith', '2019-03-22', 'SA_REP', 55000, 0.150, 'E011', 'D02'), ('E003', 'Michael', 'Johnson', '2021-06-10', 'IT_PROG', 62000, NULL, 'E010', 'D01'), ('E004', 'Emily', 'Davis', '2018-11-05', 'SA_REP', 58000, 0.200, 'E011', 'D02'), ('E005', 'David', 'Wilson', '2022-02-14', 'IT_PROG', 60000, NULL, 'E010', 'D01'), ('E006', 'Sarah', 'Brown', '2020-07-30', 'HR_REP', 50000, 0.150, 'E012', 'D03'), ('E007', 'James', 'Taylor', '2019-09-18', 'SA_REP', 60000, 0.180, 'E011', 'D02'), ('E008', 'Lisa', 'Anderson', '2021-04-25', 'FI_ACC', 52000, NULL, 'E013', 'D04'), ('E009', 'Robert', 'Martinez', '2020-12-01', 'IT_PROG', 67000, NULL, 'E010', 'D01'), ('E010', 'Patricia', 'Garcia', '2017-05-10', 'IT_MAN', 85000, NULL, NULL, 'D01'), ('E011', 'Thomas', 'Lee', '2016-08-20', 'SA_MAN', 80000, 0.100, NULL, 'D02'), ('E012', 'Maria', 'White', '2018-02-15', 'HR_MAN', 75000, NULL, NULL, 'D03'), ('E013', 'Daniel', 'Harris', '2019-10-30', 'FI_MAN', 78000, NULL, NULL, 'D04'), ('E014', 'Jessica', 'Clark', '2021-01-20', 'HR_REP', 48000, 0.100, 'E012', 'D03'), ('E015', 'Kevin', 'Lewis', '2022-03-12', 'FI_ACC', 51000, NULL, 'E013', 'D04');
No 2 SS (of Query & Output in Shell)	<pre> MariaDB [Company_23201376]> INSERT INTO employees (e_id, first_name, last_name, hire_date, job_id, salary, commission_pct, manager_id, dept_id) VALUES -> ('E001', 'John', 'Doe', '2020-01-15', 'IT_PROG', 65000, NULL, 'E010', 'D01'), -> ('E002', 'Jane', 'Smith', '2019-03-22', 'SA_REP', 55000, 0.150, 'E011', 'D02'), -> ('E003', 'Michael', 'Johnson', '2021-06-10', 'IT_PROG', 62000, NULL, 'E010', 'D01'), -> ('E004', 'Emily', 'Davis', '2018-11-05', 'SA_REP', 58000, 0.200, 'E011', 'D02'), -> ('E005', 'David', 'Wilson', '2022-02-14', 'IT_PROG', 60000, NULL, 'E010', 'D01'), -> ('E006', 'Sarah', 'Brown', '2020-07-30', 'HR_REP', 50000, 0.150, 'E012', 'D03'), -> ('E007', 'James', 'Taylor', '2019-09-18', 'SA_REP', 60000, 0.180, 'E011', 'D02'), -> ('E008', 'Lisa', 'Anderson', '2021-04-25', 'FI_ACC', 52000, NULL, 'E013', 'D04'), -> ('E009', 'Robert', 'Martinez', '2020-12-01', 'IT_PROG', 67000, NULL, 'E010', 'D01'), -> ('E010', 'Patricia', 'Garcia', '2017-05-10', 'IT_MAN', 85000, NULL, NULL, 'D01'), -> ('E011', 'Thomas', 'Lee', '2016-08-20', 'SA_MAN', 80000, 0.100, NULL, 'D02'), -> ('E012', 'Maria', 'White', '2018-02-15', 'HR_MAN', 75000, NULL, NULL, 'D03'), -> ('E013', 'Daniel', 'Harris', '2019-10-30', 'FI_MAN', 78000, NULL, NULL, 'D04'), -> ('E014', 'Jessica', 'Clark', '2021-01-20', 'HR_REP', 48000, 0.100, 'E012', 'D03'), -> ('E015', 'Kevin', 'Lewis', '2022-03-12', 'FI_ACC', 51000, NULL, 'E013', 'D04'); Query OK, 15 rows affected (0.263 sec) Records: 15 Duplicates: 0 Warnings: 0 </pre>
No 3 Query (as Plain Text)	SELECT first_name, last_name, hire_date, dept_id FROM employees

	WHERE hire_date = (SELECT MAX(hire_date) FROM employees);
No 3 SS (of Query & Output in Shell)	<pre> MariaDB [Company_23201376]> SELECT first_name, last_name, hire_date, dept_id -> FROM employees -> WHERE hire_date = (SELECT MAX(hire_date) FROM employees); +-----+-----+-----+-----+ first_name last_name hire_date dept_id +-----+-----+-----+-----+ Kevin Lewis 2022-03-12 D04 +-----+-----+-----+-----+ 1 row in set (0.042 sec) </pre>
No 4 Query (as Plain Text)	SELECT first_name, last_name, e_id, salary, dept_id FROM employees WHERE (dept_id, salary) IN (SELECT dept_id, MAX(salary) FROM employees GROUP BY dept_id);
No 4 SS (of Query & Output in Shell)	<pre> MariaDB [Company_23201376]> SELECT first_name, last_name, e_id, salary, dept_id FROM employees WHERE (dept_id, salary) IN (SELECT dept_id, MAX(salary) FROM employees GR OUP BY dept_id); +-----+-----+-----+-----+-----+ first_name last_name e_id salary dept_id +-----+-----+-----+-----+-----+ Patricia Garcia E010 85000 D01 Thomas Lee E011 80000 D02 Maria White E012 75000 D03 Daniel Harris E013 78000 D04 +-----+-----+-----+-----+-----+ 4 rows in set (0.094 sec) </pre>
No 5 Query (as Plain Text)	SELECT first_name, last_name, e_id, commission_pct, dept_id FROM employees WHERE dept_id = 'D02' AND commission_pct > ALL (SELECT commission_pct FROM employees WHERE dept_id = 'D01' AND commission_pct IS NOT NULL);

No 5 SS (of Query & Output in Shell)	<pre>MariaDB [Company_23201376]> SELECT first_name, last_name, e_id, commission_pct, dept_id FROM employees WHERE dept_id = 'D02' AND commission_pct > ALL (SELECT commission_pct FROM employees WHERE dept_id = 'D01' AND commission_pct IS NOT NULL);</pre> <table><thead><tr><th>first_name</th><th>last_name</th><th>e_id</th><th>commission_pct</th><th>dept_id</th></tr></thead><tbody><tr><td>Jane</td><td>Smith</td><td>E002</td><td>0.150</td><td>D02</td></tr><tr><td>Emily</td><td>Davis</td><td>E004</td><td>0.200</td><td>D02</td></tr><tr><td>James</td><td>Taylor</td><td>E007</td><td>0.180</td><td>D02</td></tr><tr><td>Thomas</td><td>Lee</td><td>E011</td><td>0.100</td><td>D02</td></tr></tbody></table> <pre>4 rows in set (0.086 sec)</pre>	first_name	last_name	e_id	commission_pct	dept_id	Jane	Smith	E002	0.150	D02	Emily	Davis	E004	0.200	D02	James	Taylor	E007	0.180	D02	Thomas	Lee	E011	0.100	D02
first_name	last_name	e_id	commission_pct	dept_id																						
Jane	Smith	E002	0.150	D02																						
Emily	Davis	E004	0.200	D02																						
James	Taylor	E007	0.180	D02																						
Thomas	Lee	E011	0.100	D02																						
No 6 Query (as Plain Text)	SELECT dept_id, COUNT(e_id) AS total_employees FROM employees GROUP BY dept_id HAVING MAX(salary) <= 70000;																									
No 6 SS (of Query & Output in Shell)	<pre>MariaDB [Company_23201376]> SELECT dept_id, COUNT(e_id) AS total_employees FROM employees GROUP BY dept_id HAVING MAX(salary) <= 70000; Empty set (0.037 sec)</pre>																									
No 7 Query (as Plain Text)	SELECT dept_id, job_id, salary -> FROM employees e1 -> WHERE salary < ANY (SELECT salary FROM employees e2 WHERE e1.dept_id = e2.dept_id AND e1.job_id <> e2.job_id);																									

No 7 SS (of Query & Output in Shell)	<pre> MariaDB [Company_23201376]> SELECT dept_id, job_id, salary -> FROM employees e1 -> WHERE salary < ANY (SELECT salary FROM employees e2 WHERE e1.dept_id = e2.dept_id AND e1.job_id <> e2.job_id); +-----+-----+-----+ dept_id job_id salary +-----+-----+-----+ D01 IT_PROG 65000 D02 SA_REP 55000 D01 IT_PROG 62000 D02 SA_REP 58000 D01 IT_PROG 60000 D03 HR_REP 50000 D02 SA_REP 60000 D04 FI_ACC 52000 D01 IT_PROG 67000 D03 HR_REP 48000 D04 FI_ACC 51000 +-----+-----+-----+ 11 rows in set (0.111 sec) </pre>
No 8 Query (as Plain Text)	SELECT DISTINCT manager_id FROM employees WHERE manager_id IS NOT NULL AND manager_id NOT IN (SELECT manager_id FROM employees WHERE salary < 55000 AND manager_id IS NOT NULL);
No 8 SS (of Query & Output in Shell)	<pre> MariaDB [Company_23201376]> SELECT DISTINCT manager_id FROM employees WHERE manager_id IS NOT NULL AND manager_id NOT IN (SELECT manager_id FROM employees WHERE salary < 55000 AND manager_id IS NOT NULL); +-----+ manager_id +-----+ E010 E011 +-----+ 2 rows in set (0.005 sec) </pre>
No 9 Query (as Plain Text)	SELECT first_name, last_name, e_id, salary, dept_id, commission_pct FROM employees WHERE (manager_id, commission_pct) IN (SELECT manager_id, MAX(commission_pct) FROM employees WHERE manager_id IS NOT NULL GROUP BY manager_id);

No 9 SS
(of Query & Output
in Shell)

```
MariaDB [Company_23201376]> SELECT first_name, last_name, e_id, salary, dept_id, commission_pct FROM employees WHERE (manager_id, commission_pct) IN (SELECT manager_id,
MAX(commission_pct) FROM employees WHERE manager_id IS NOT NULL GROUP BY manager_id);
+-----+-----+-----+-----+-----+-----+
| first_name | last_name | e_id | salary | dept_id | commission_pct |
+-----+-----+-----+-----+-----+-----+
| Emily      | Davis     | E004 | 58000  | D02     | 0.200          |
| Sarah      | Brown     | E006 | 50000  | D03     | 0.150          |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.032 sec)
```

No 10 Query
(as Plain Text)

SELECT job_id, AVG(salary) AS avg_job_salary FROM employees GROUP BY job_id HAVING AVG(salary) < (SELECT
AVG(salary) FROM employees);

No 10 SS
(of Query & Output
in Shell)

```
MariaDB [Company_23201376]> SELECT job_id, AVG(salary) AS avg_job_salary FROM employees GROUP BY job_id HAVING AVG(salary) < (SELECT AVG(salary) FROM employees);
+-----+-----+
| job_id | avg_job_salary |
+-----+-----+
| FI_ACC | 51500.0000     |
| HR_REP | 49000.0000     |
| SA_REP | 57666.6667     |
+-----+-----+
3 rows in set (0.020 sec)
```